

Electronics Homework 2 : BJT Amplifier Design

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Design Specifications and Goals:

- $V_{CC} = 15V$, $R_{sig} = 2k\Omega$, $R_L = 10k\Omega$.
- Transistor: 2N3904 (NPN).
- Targets: Gain $|v_o/v_{sig}| \geq 50$, output swing $-3V \leq v_o \leq 3V$, and power consumption $< 150mW$.

Problem a (*Design of bias current and gain derivation*)

1. Derivation of Equation (1)

Based on the small-signal model analysis and assuming $R_{in} \cong r_\pi$:

- **Base voltage v_b :** Applying the voltage divider rule at the input stage:

$$v_b = v_{sig} \frac{r_\pi}{R_{sig} + r_\pi} = v_{sig} \frac{\beta V_T / I_C}{R_{sig} + \beta V_T / I_C} = v_{sig} \frac{\beta V_T}{\beta V_T + I_C R_{sig}}$$

- **Output voltage v_o :** $v_o = -g_m v_b (R_o \parallel R_L)$, where $g_m = I_C / V_T$ and $R_o \cong R_C$.
- **Substitution:** Substituting v_b and using $V_{RC} = I_C R_C \implies R_C = V_{RC} / I_C$:

$$\frac{v_o}{v_{sig}} = -\frac{\beta I_C}{\beta V_T + I_C R_{sig}} \times \frac{V_{RC} R_L}{V_{RC} + R_L I_C} \quad (1)$$

2. Selection of Bias Current I_C

Following the experience rule, we set the voltage across R_C as $V_{RC} = V_{CC}/3 = 5V$. To determine the optimal bias current, we evaluate the voltage gain for the two operating points provided in Table 1:

• Case 1: $I_C = 0.5mA$

- From Table 1, $\beta \approx 303.714$.
- Substituting into Equation (1) with $V_T = 25mV$, $R_{sig} = 2k\Omega$, and $R_L = 10k\Omega$:

$$|A_v| \approx \left| -\frac{303.714 \cdot 0.5m}{303.714 \cdot 25m + 0.5m \cdot 2k} \cdot \frac{5 \cdot 10k}{5 + 0.5m \cdot 10k} \right| \approx 88.3624$$

– This satisfies the design requirement of $|v_o/v_{sig}| \geq 50$.

• **Case 2: $I_C = 10\text{mA}$**

– From Table 1, $\beta \approx 296.53$.

– Substituting into Equation (1) with $V_T = 25\text{mV}$, $R_{sig} = 2\text{k}\Omega$, and $R_L = 10\text{k}\Omega$:

$$|A_v| \approx \left| -\frac{296.53 \cdot 10\text{m}}{296.53 \cdot 25\text{m} + 10\text{m} \cdot 2\text{k}} \cdot \frac{5 \cdot 10\text{k}}{5 + 10\text{m} \cdot 10\text{k}} \right| \approx 51.5097$$

– This satisfies the design requirement of $|v_o/v_{sig}| \geq 50$.

Conclusion:

After evaluating both operating points, we select $I_C = 0.5\text{mA}$ as the desired bias current for the following reasons:

- **Higher Gain Margin:** The calculated gain for $I_C = 0.5\text{mA}$ is approximately **88.36**, which provides a much larger safety margin above the required threshold of $|v_o/v_{sig}| \geq 50$ compared to the 10mA case (≈ 51.51).
- **Power Efficiency:** By operating at a significantly lower collector current, we ensure that the total power dissipation remains well below the strict limit of **150mW**. This minimizes thermal stress on the 2N3904 transistor and complies with the design specifications.

Problem b (*Design of Resistors for $I_C = 0.5\text{mA}$*)

Resistor Calculations based on Experience Rules

Following the specific hints provided in the assignment, we design the resistors to bias the BJT at $I_C = 0.5\text{mA}$ with $V_{CC} = 15\text{V}$.

- **Collector Resistor (R_C):** Using the rule $V_{RC} = V_{CC}/3 = 5\text{V}$:

$$R_C = \frac{V_{RC}}{I_C} = \frac{5\text{V}}{0.5\text{mA}} = \mathbf{10\text{k}\Omega}.$$

- **Emitter Resistor (R_E):** Given $V_{BB} \cong V_{CC}/3 = 5\text{V}$ and $V_{BE} \cong 0.7\text{V}$:

$$V_E = V_{BB} - V_{BE} = 5\text{V} - 0.7\text{V} = 4.3\text{V}.$$

Since $I_E \approx I_C = 0.5\text{mA}$:

$$R_E = \frac{V_E}{I_E} = \frac{4.3\text{V}}{0.5\text{mA}} = \mathbf{8.6\text{k}\Omega}.$$

- **Base Voltage Divider (R_1, R_2):** According to the hint $I_{R1} = I_{R2} \cong 0.1I_E$:

$$I_{R1} = I_{R2} = 0.1 \times 0.5\text{mA} = 50\mu\text{A}.$$

Applying Ohm's Law for the divider network:

- $R_2 = \frac{V_{BB}}{I_{R2}} = \frac{5V}{50\mu A} = 100k\Omega.$
- $R_1 = \frac{V_{CC}-V_{BB}}{I_{R1}} = \frac{15V-5V}{50\mu A} = \frac{10V}{50\mu A} = 200k\Omega.$

Problem c (Simulation in LTspice - Performance Analysis)

1. Circuit Schematic

The complete circuit schematic, including simulation commands (.op and .tran 10m), is shown in Figure 1. All resistor values were set as $R_1 = 200k\Omega$, $R_2 = 100k\Omega$, $R_C = 10k\Omega$, and $R_E = 8.6k\Omega$.

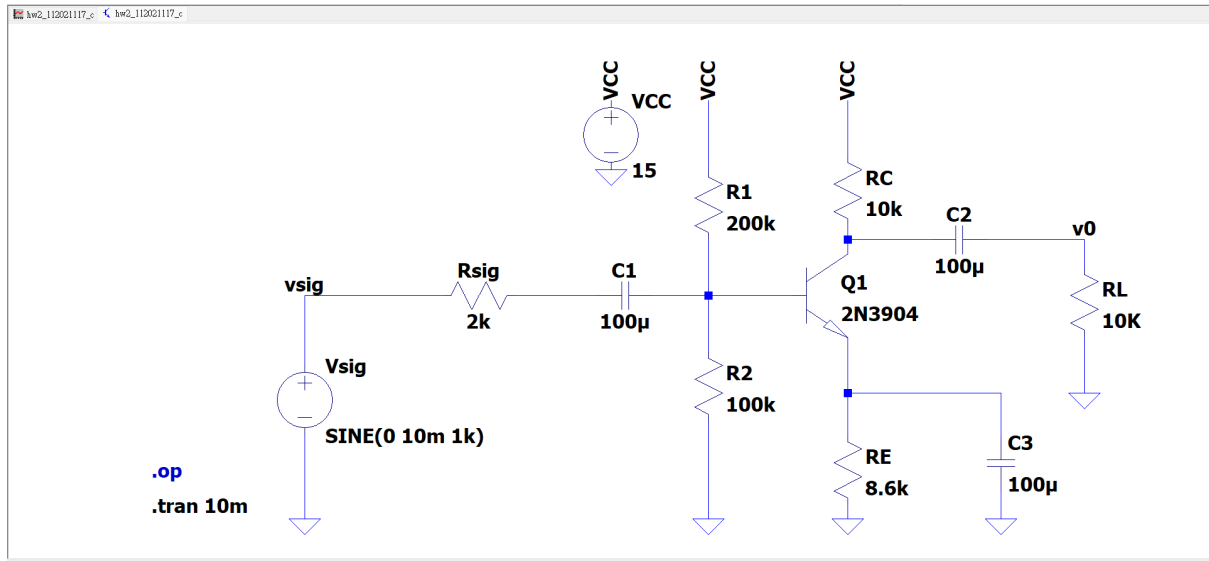


Figure 1: LTspice schematic of the Common-Emitter amplifier.

2. Power Dissipation Analysis

According to the DC operating point analysis (Figure 2), the current supplied by V_{CC} is $|I(V_{CC})| = 0.544mA$. The total power consumption is:

$$P_{diss} = 15V \times 0.544mA = 8.16mW.$$

Conclusion: The power dissipation (8.16 mW) is well below the 150mW limit.

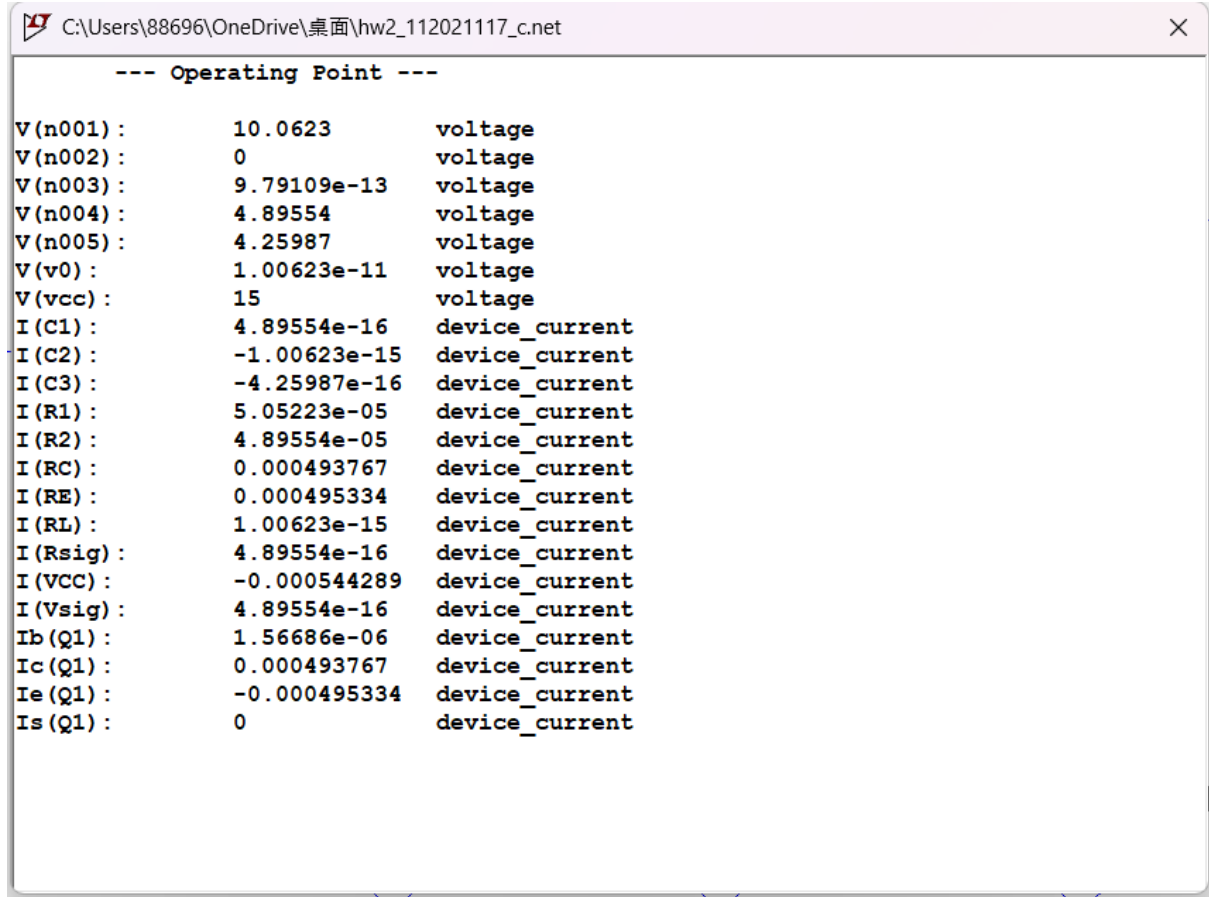


Figure 2: DC operating point analysis results.

3. Voltage Gain Verification

The voltage gain is verified by measuring the peak-to-peak values of the input and output waveforms using cursors (Figure 3 and Figure 4):

- **Input Signal ($V_{sig,pp}$):** Measured as 19.99mV (≈ 20 mV).
- **Output Signal ($V_{o,pp}$):** Measured as 1.602V.
- **Measured Voltage Gain (A_v):**

$$|A_v| = \frac{V_{o,pp}}{V_{sig,pp}} = \frac{1.602V}{0.01999V} \approx 80.14.$$

Conclusion: The simulated gain of **80.14** is well above the required threshold of 50.

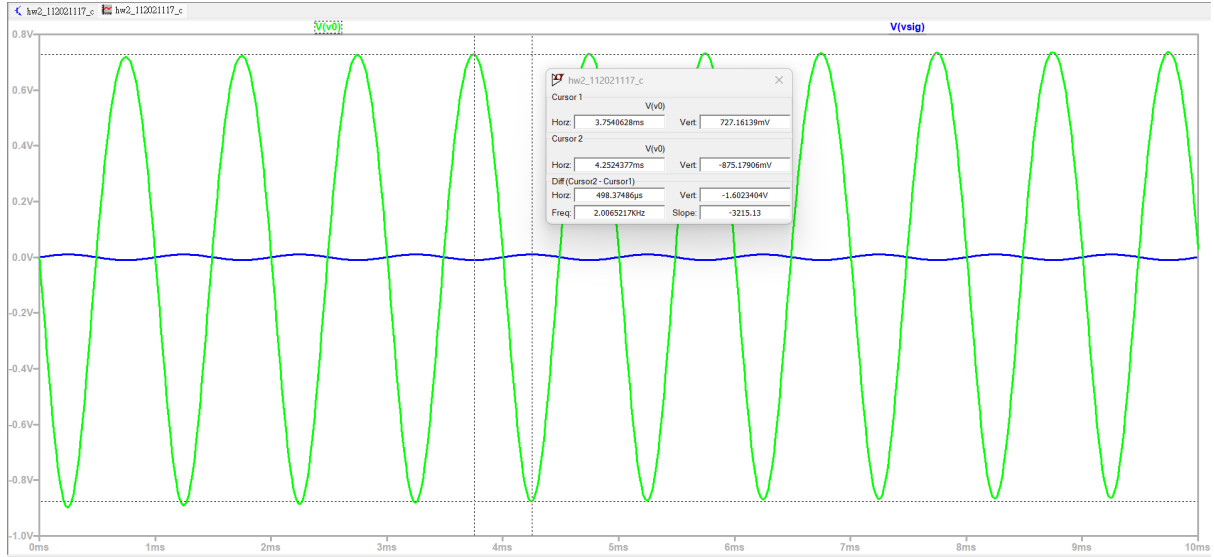


Figure 3: Output waveform v_o showing a peak-to-peak value of 1.602V.

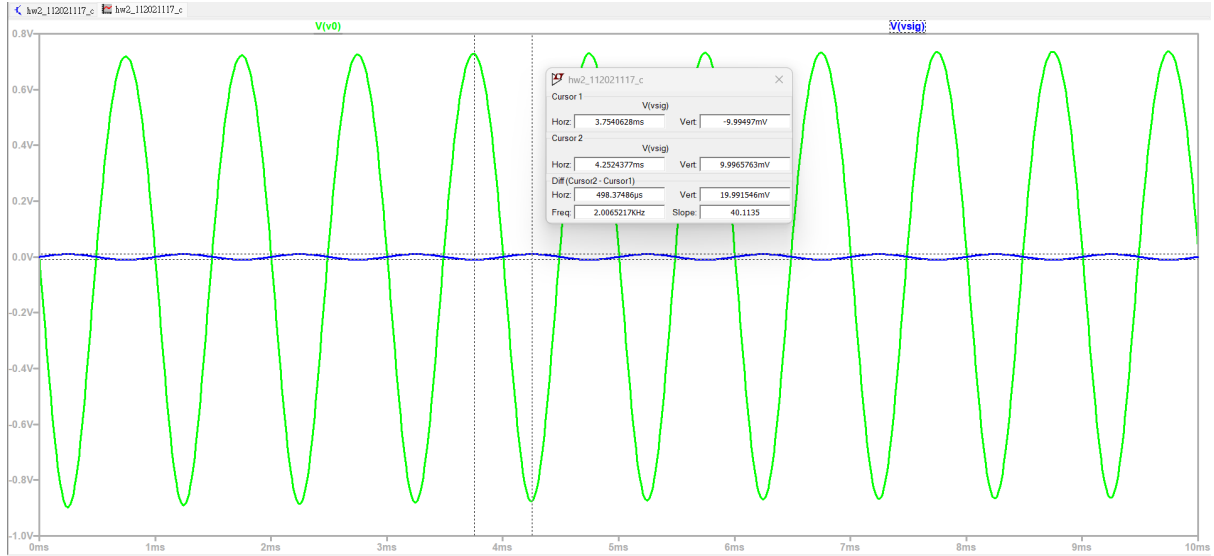


Figure 4: Input waveform v_{sig} showing a peak-to-peak value of 19.99mV.

Problem d (*Large Signal Analysis and Resistor Fine-Tuning*)

1. Initial Failure Analysis

The original design with $R_C = 10\text{ k}\Omega$ and $I_C = 0.5\text{ mA}$ was tested with a 100 mV amplitude input signal. The output waveform v_o experienced asymmetric clipping (as shown in Figure 5) because the positive output swing is physically limited by the DC bias:

$$v_{o,\max} = I_C \cdot (R_C \parallel R_L) = 0.5\text{ mA} \times \frac{10\text{ k} \times 10\text{ k}}{10\text{ k} + 10\text{ k}} = 0.5\text{ mA} \times 5\text{ k}\Omega = 2.5\text{ V}$$

Since $2.5\text{ V} < 3\text{ V}$, the initial design cannot satisfy the required output range of $\pm 3\text{ V}$.

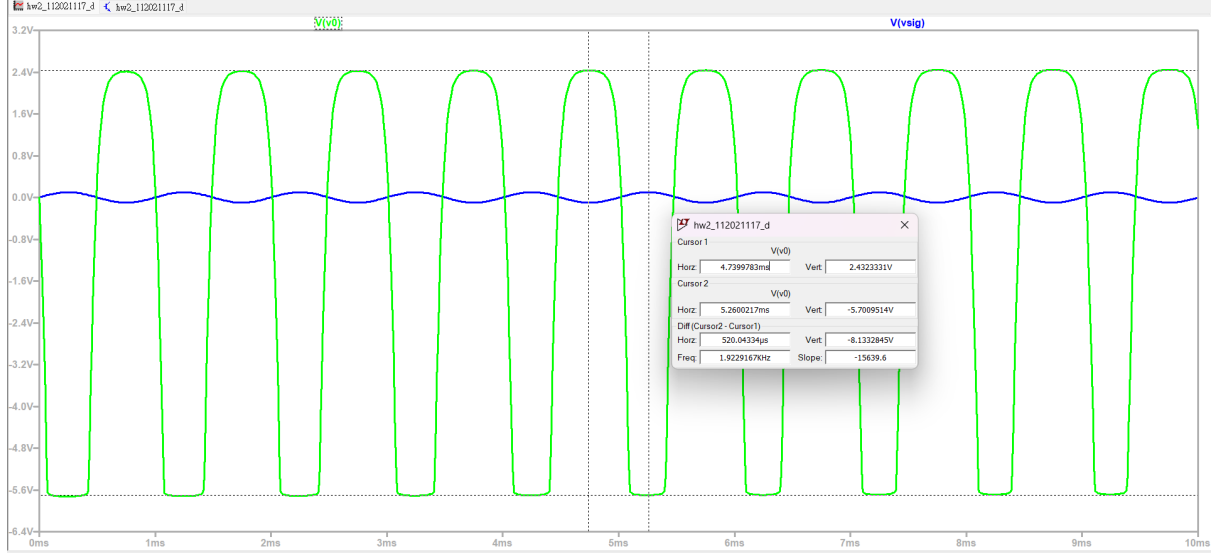


Figure 5: Initial simulation with $R_C = 10\text{k}\Omega$ showing severe clipping at 100mV input.

2. Fine-Tuning Strategy

Why only R_C is adjusted: The bias current I_C is determined solely by R_1 , R_2 , and R_E :

$$V_{BB} = V_{CC} \cdot \frac{R_2}{R_1 + R_2} = \text{constant}, \quad I_C \approx \frac{V_{BB} - V_{BE}}{R_E} = \text{constant}$$

As long as R_1 , R_2 , and R_E remain unchanged, I_C stays fixed at 0.5 mA. Therefore, only R_C needs to be adjusted to relocate the Q-point.

Effect of increasing R_C : From KVL around the collector loop:

$$V_{CC} = \underbrace{I_C R_C}_{V_{RC}} + \underbrace{V_{CE,Q}}_{\text{Q-point}} + \underbrace{I_E R_E}_{V_{RE}}$$

Since V_{CC} and V_{RE} are fixed, V_{RC} and $V_{CE,Q}$ are inversely related. Increasing R_C produces two simultaneous effects:

- $R_C \uparrow \Rightarrow (R_C \parallel R_L) \uparrow \Rightarrow$ **positive swing increases**
- $R_C \uparrow \Rightarrow V_{CE,Q} \downarrow \Rightarrow$ **negative swing decreases**

This simultaneously corrects the insufficient positive swing and the excessive negative swing of the original design.

3. Calculation of Required R_C and Final Schematic

For symmetric clipping at $\pm 3\text{V}$, we require:

$$V_{CE,Q} - V_{CE,sat} = 3\text{V} \Rightarrow V_{CE,Q} = 3.2\text{V}$$

Applying KVL with $V_{RE} = V_{BB} - V_{BE} = 5 - 0.7 = 4.3\text{V}$:

$$V_{RC} = V_{CC} - V_{CE,Q} - V_{RE} = 15 - 3.2 - 4.3 = 7.5\text{V} \Rightarrow R_C = \frac{7.5\text{V}}{0.5\text{mA}} = 15\text{k}\Omega$$

After simulation, R_C was further fine-tuned to $16\text{ k}\Omega$ to provide additional margin. The final schematic is shown in Figure 6.

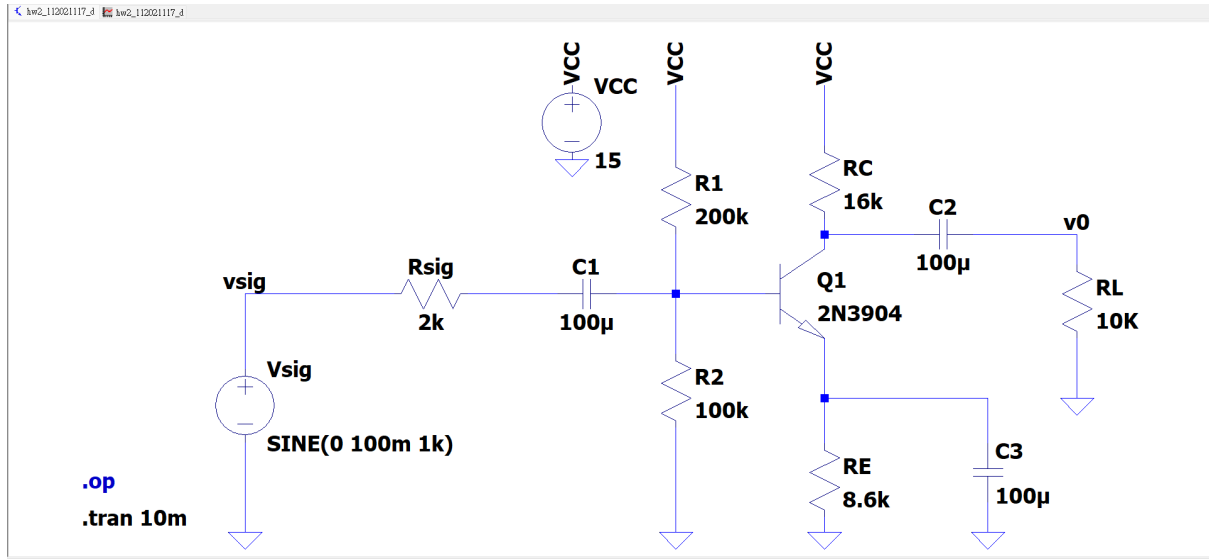


Figure 6: Final optimized schematic with $R_C = 16\text{ k}\Omega$.

4. Simulation Results and Power Verification

The DC operating point analysis (Figure 7) shows $I(V_{CC}) = -0.5438\text{ mA}$.

$$P = V_{CC} \times |I(V_{CC})| = 15\text{ V} \times 0.5438\text{ mA} = 8.157\text{ mW} \ll 150\text{ mW} \checkmark$$

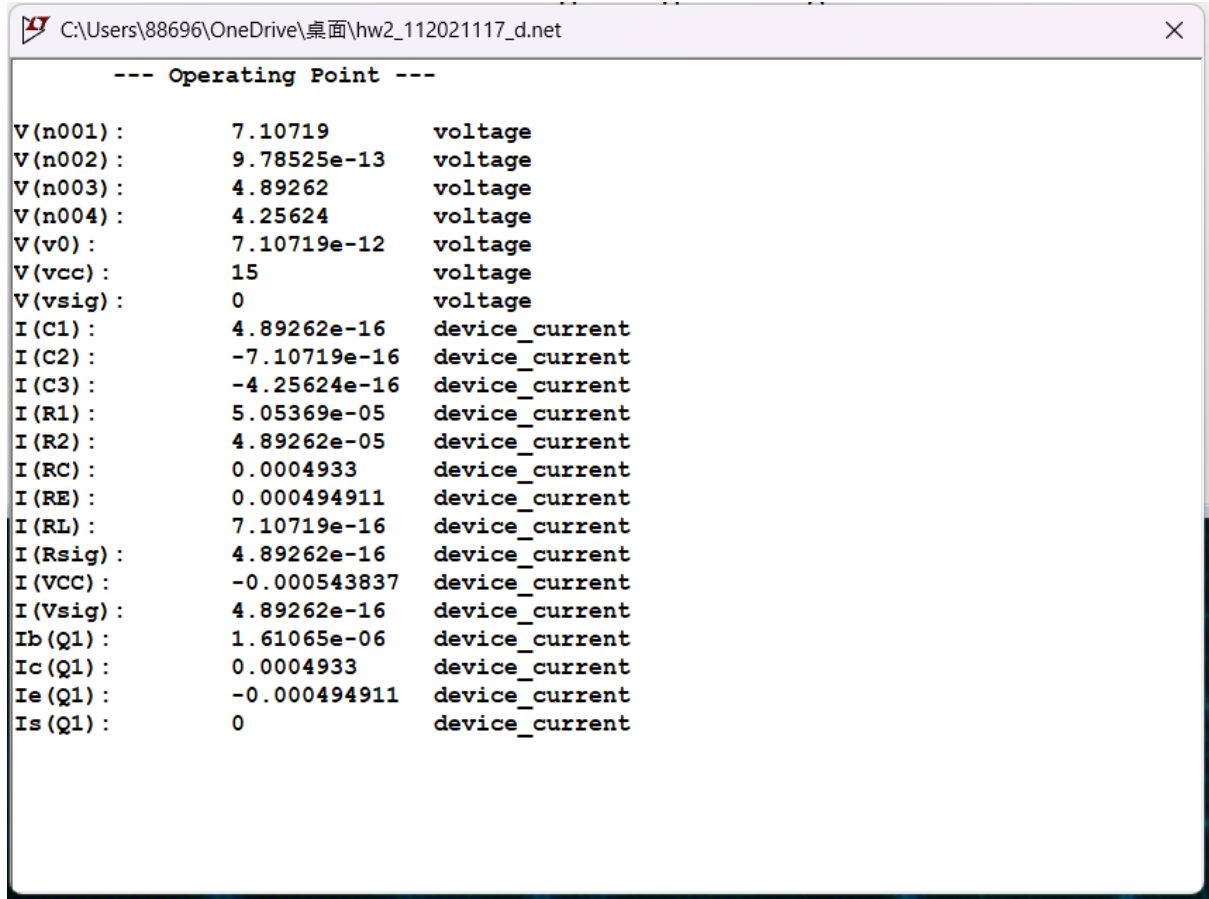


Figure 7: DC operating point results for the 16kΩ design.

The final measured output swing $V_{o,pp}$ is **5.753V** (Figure 8), confirming that the waveform stays within $-3\text{ V} \leq v_o \leq 3\text{ V}$. The input $V_{sig,pp}$ was verified as **199.8mV** (Figure 9).

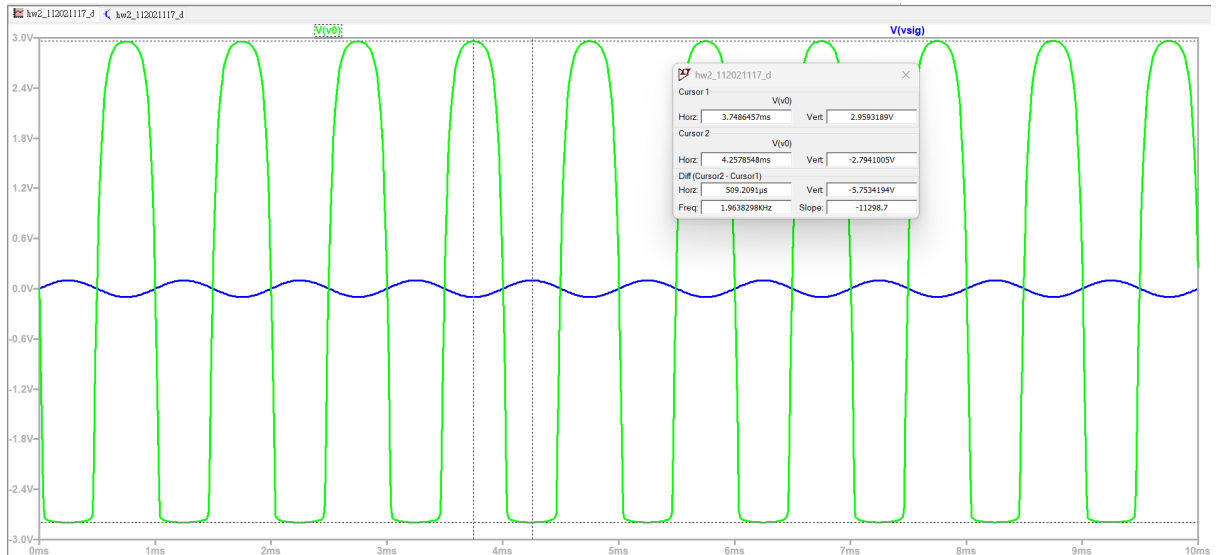


Figure 8: Output waveform v_o with measured peak-to-peak swing of 5.753V.

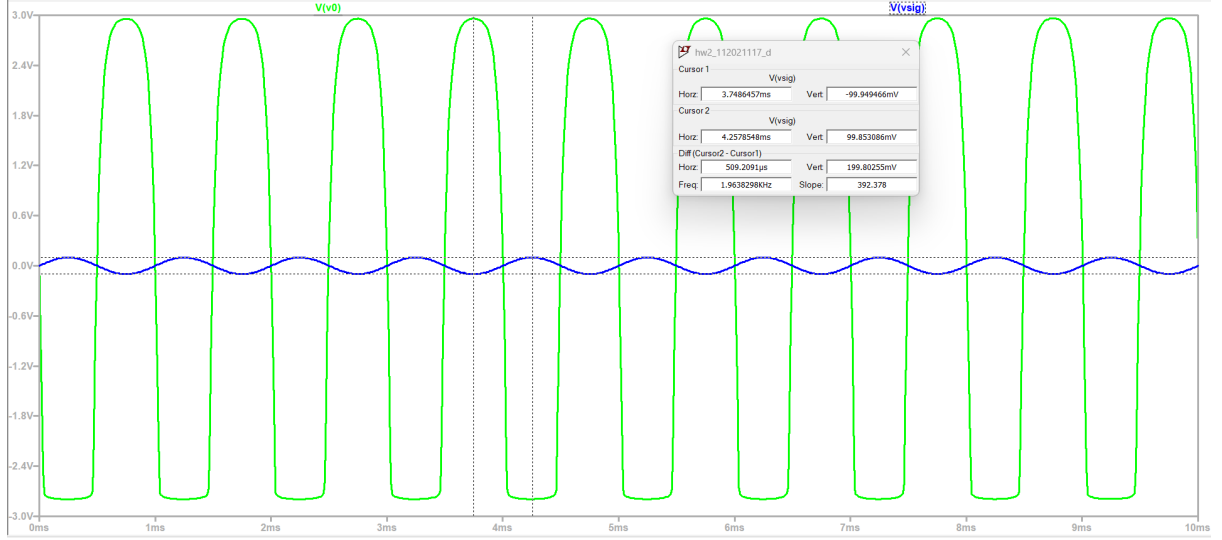


Figure 9: Input waveform v_{sig} measured at 199.8mV peak-to-peak.

5. Does the mean value of v_o equal zero?

In principle, the AC coupling capacitor C_2 blocks the DC component of v_o , so the mean value should be zero. However, since the positive and negative clipping levels are not perfectly symmetric due to BJT nonlinearity, the waveform is slightly asymmetric. As a result, the mean value of v_o is **not exactly zero**, though the deviation is small.

Problem e (*Gain Reduction via Emitter Degeneration*)

1. Systematic Derivation of the Gain Equation

To analyze the effect of R_3 , we define the effective emitter impedance as $Z_E = R_E \parallel (R_3 + Z_{C3})$. Since $R_E \gg R_3$ and C_3 acts as a short circuit at 1 kHz, $Z_E \approx R_3$. The total voltage gain is derived from the generalized small-signal model:

- **Input Impedance (R_{in}):** $R_{in} = r_\pi + (\beta + 1)Z_E$.
- **Base-to-Signal Ratio:** Based on the input voltage divider:

$$\frac{v_b}{v_{sig}} = \frac{R_{in}}{R_{sig} + R_{in}} = \frac{r_\pi + (\beta + 1)Z_E}{R_{sig} + r_\pi + (\beta + 1)Z_E}$$

- **Output-to-Base Ratio:** The gain from base to collector is:

$$\frac{v_o}{v_b} = -\frac{\beta(R_C \parallel R_L)}{R_{in}} = -\frac{\beta(R_C \parallel R_L)}{r_\pi + (\beta + 1)Z_E}$$

- **Generalized Gain Formula:** Combining the terms, we obtain:

$$A_v = \frac{v_o}{v_{sig}} = -\frac{\beta(R_C \parallel R_L)}{R_{sig} + r_\pi + (\beta + 1)Z_E} \quad (2)$$

2. Theoretical Calculation of R_3

Let $|A_{v,old}|$ be the gain when $Z_E \approx 0$ (from Problem c), and $|A_{v,new}|$ be the target gain where $Z_E \approx R_3$. We require:

$$\frac{|A_{v,old}|}{|A_{v,new}|} = \frac{R_{sig} + r_\pi + (\beta + 1)R_3}{R_{sig} + r_\pi} = 3 \implies R_3 = \frac{2(R_{sig} + r_\pi)}{\beta + 1}$$

Using $R_{sig} = 2\text{ k}\Omega$, $r_\pi = 15.18\text{ k}\Omega$ (for $I_C = 0.5\text{ mA}$), and $\beta = 303.7$:

$$R_3 = \frac{2(2\text{ k}\Omega + 15.18\text{ k}\Omega)}{304.7} \approx \mathbf{112.8\Omega}$$

3. Parameter Calibration and Reverse Engineering

An initial simulation was performed with the theoretical value $R_3 = 112.8\Omega$. As shown in Figure 10, the measured output $V_{o,pp}$ was **691.25 mV**. This corresponds to a gain reduction ratio of 2.317 ($1.602\text{ V}/0.691\text{ V}$), which deviates from the target ratio of 3. This discrepancy is primarily due to the **Early Effect** and dynamic β variations at the specific biasing point.

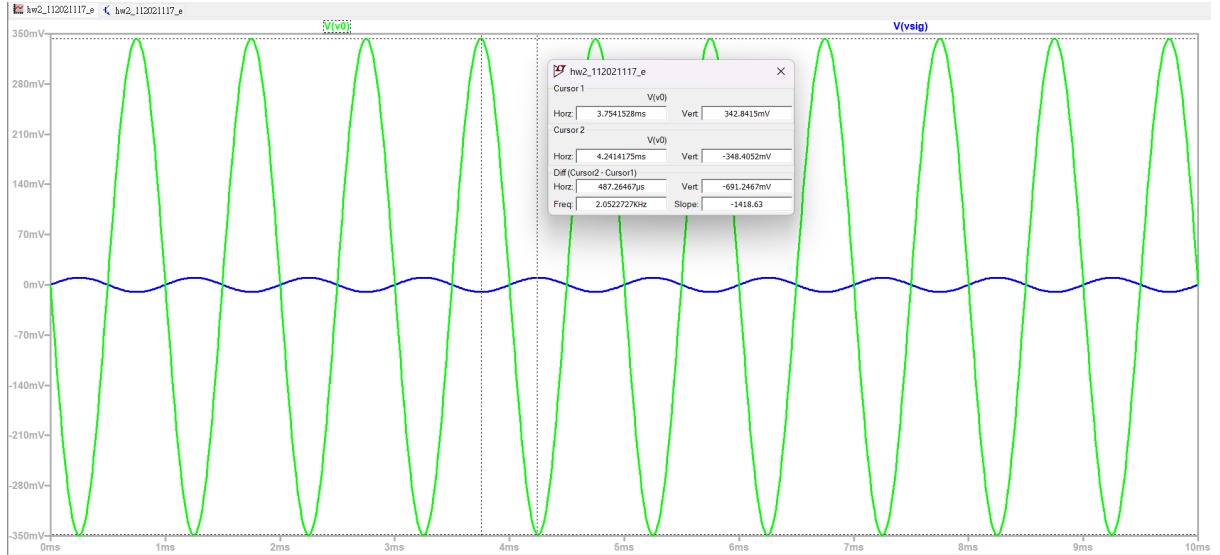


Figure 10: Calibration simulation with theoretical $R_3 = 112.8\Omega$ showing $V_{o,pp} = 691.25\text{ mV}$.

To correct this, we treat the term $\frac{R_{sig} + r_\pi}{\beta + 1}$ as a circuit constant. From the calibration trial:

$$\text{Ratio} = 1 + \frac{R_3}{\frac{R_{sig} + r_\pi}{\beta + 1}} \implies \frac{R_{sig} + r_\pi}{\beta + 1} = \frac{112.8\Omega}{2.317 - 1} \approx 85.6\Omega$$

For a target ratio of 3:

$$R_{3,target} = (3 - 1) \times 85.6\Omega \approx \mathbf{171.2\Omega}$$

4. Final Verification

The final optimized schematic with $R_3 = 171.2\ \Omega$ is shown in Figure 11. The input signal was verified to be a stable 19.99 mV_{pp} (Figure 12), and the resulting output $V_{o,pp}$ reached **529.4 mV** (Figure 13), perfectly satisfying the required 1/3 gain reduction goal.

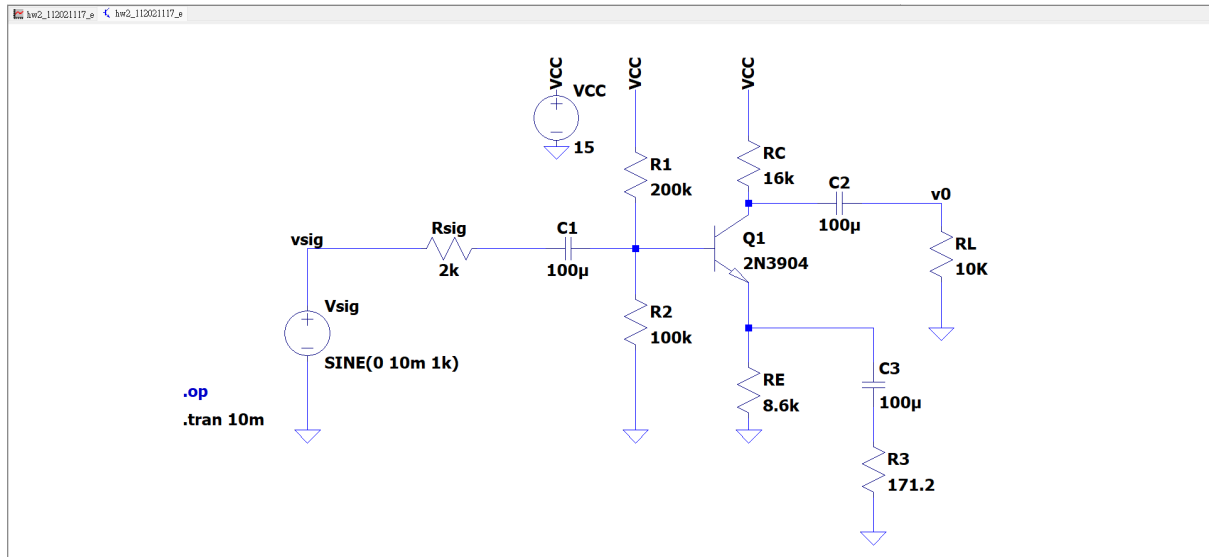


Figure 11: Final optimized schematic with $R_3 = 171.2\ \Omega$.

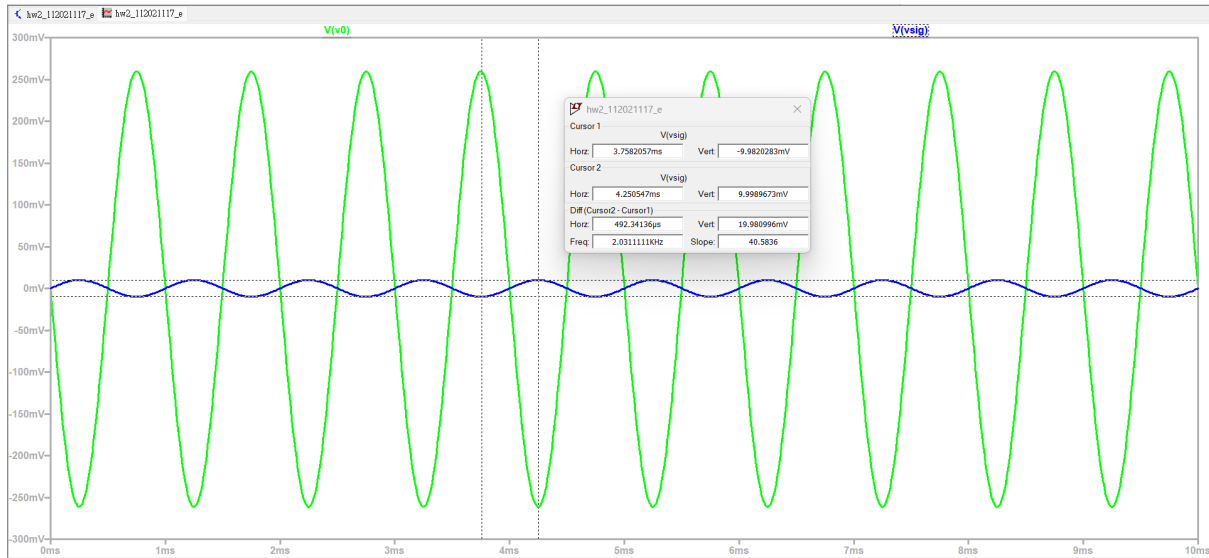


Figure 12: Input signal verification: $V_{sig,pp} = 19.99\text{ mV}$.

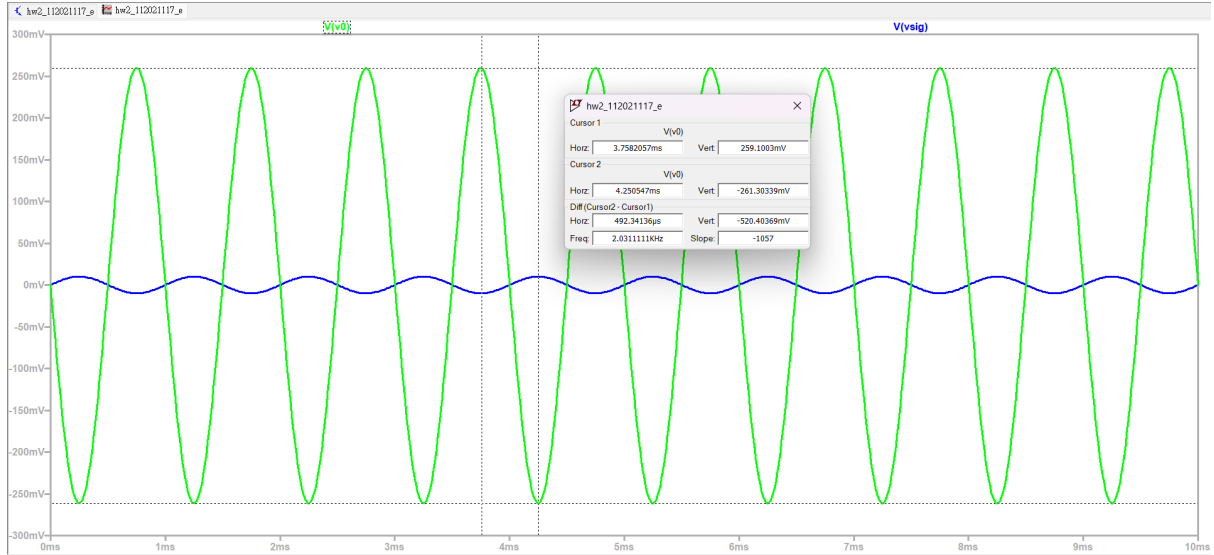


Figure 13: Final output verification: $V_{o,pp} = 529.4\text{mV}$ (Gain ≈ 26.5).

Problem f (Design of Emitter Follower Stage for Low-Impedance Load)

1. Problem Analysis and Impedance Isolation

Directly connecting a reduced load of $R_L = 100\Omega$ to the CE amplifier drastically drops the effective AC load to $R_{L,\text{eff}} = R_C \parallel R_L \approx 99.4\Omega$, causing the voltage gain to plummet to $|A_v| \approx 2$. To preserve a total gain $|v_o/v_{sig}| \geq 50$, an Emitter Follower buffer stage must be inserted to isolate the low load impedance. The boundary limits for the design parameters I_1 and R_B are derived from scratch below.

2. Derivation of Bias Current I_1 (Large-Signal Swing Constraint)

To prevent Q_1 from entering the cutoff region during negative signal peaks, the total instantaneous emitter current $i_{E1}(t) = I_1 + i_L(t)$ must remain strictly positive ($i_{E1}(t) > 0$).

- **Peak AC Load Current:** For the required range $-3\text{V} \leq v_o \leq 3\text{V}$ and $R_L = 100\Omega$:

$$i_{L,\text{peak}} = \frac{v_{o,\text{min}}}{R_L} = \frac{-3\text{V}}{100\Omega} = -30\text{mA}$$

- **Minimum Bound for I_1 :** Bounding $i_{E1}(t) > 0$ at the negative peak yields:

$$I_1 + i_{L,\text{peak}} > 0 \implies I_1 > 30\text{mA}$$

- **Design Choice:** Applying a safety margin factor of 1.6 to ensure linearity near the negative rail, we select $I_1 = 50\text{mA}$.

3. Derivation of Base Resistor R_B (AC Gain and DC Bias Trade-off)

The selection of R_B is bounded by the minimum voltage gain requirement (AC) and the Q-point linearity limit (DC). Assuming a conservative large-signal $\beta_2 \approx 100$ for Q_1 :

Constraint 1: AC Voltage Gain (Lower Bound of R_B)

- The internal input impedance looking into the base of Q_1 is:

$$R_{in2, internal} = r_{\pi 2} + (\beta_2 + 1)R_L = \frac{\beta_2 V_T}{I_1} + (\beta_2 + 1)R_L = \frac{100 \times 25\text{mV}}{50\text{mA}} + 101 \times 100\Omega \approx 10.15\text{k}\Omega$$

- The stage input impedance is $R_{in2} = R_B \parallel R_{in2, internal}$. Given the follower gain is ≈ 0.96 , the CE stage gain $|A_{v1}|$ must satisfy $|A_{v1}| \geq 50/0.96 \approx 52.1$.
- Substituting the parameters ($\beta_1 = 303.7$, $R_{sig} = 2\text{k}\Omega$, $r_{\pi 1} = 15.18\text{k}\Omega$, $R_C = 16\text{k}\Omega$) into the generalized gain formula:

$$\frac{\beta_1(R_C \parallel R_{in2})}{R_{sig} + r_{\pi 1}} \geq 52.1 \implies 16\text{k}\Omega \parallel R_{in2} \geq 2.95\text{k}\Omega \implies R_{in2} \geq 3.62\text{k}\Omega$$

- Solving $R_B \parallel 10.15\text{k}\Omega \geq 3.62\text{k}\Omega$ gives the lower limit: $R_B \geq 5.62\text{k}\Omega$.

Constraint 2: DC Q-point Saturation (Upper Bound of R_B)

- The DC base voltage V_{B1} drops due to the base bias current $I_{B1} = I_1/\beta_2 = 0.5\text{mA}$:

$$V_{B1} = V_{CC} - I_{B1}R_B = 15\text{V} - (0.5\text{mA})R_B$$

- To maintain a safe collector-emitter voltage $V_{CE1,Q}$ and prevent the pre-stage collector voltage from premature clipping during a 3V peak excursion, V_{B1} must be kept above 6V:

$$15\text{V} - (0.5\text{mA})R_B \geq 6\text{V} \implies \mathbf{R_B \leq 18\text{k}\Omega}$$

Conclusion and Parameter Selection: Combining both constraints defines the allowable window: $5.62\text{k}\Omega \leq R_B \leq 18\text{k}\Omega$. A standard value of $R_B = 10\text{k}\Omega$ is selected, optimizing both the AC gain margin and DC voltage headroom.

4. Simulation and Performance Verification

The integrated multi-stage amplifier was simulated in LTspice using the derived parameters ($I_1 = 50\text{mA}$, $R_B = 10\text{k}\Omega$), as shown in Figure 14. Under a 1kHz sine wave with a 10mV amplitude, transient analysis was performed to verify the performance metrics:

- **Input Voltage Swing:** The simulated peak-to-peak input voltage $V_{sig,pp}$ is measured to be 19.95mV (Figure 15).
- **Output Voltage Swing:** The undistorted peak-to-peak output voltage $V_{o,pp}$ across the 100Ω load reaches 1.548V (Figure 16).

The overall loaded voltage gain $|A_v|$ is calculated as:

$$|A_v| = \frac{V_{o,pp}}{V_{sig,pp}} = \frac{1.548\text{V}}{0.01995\text{V}} \approx \mathbf{77.59}$$

Conclusion: The verified gain of **77.59** exceeds the design requirement of 50 V/V, proving that the emitter follower successfully isolates the heavy 100Ω load without creating distortion.

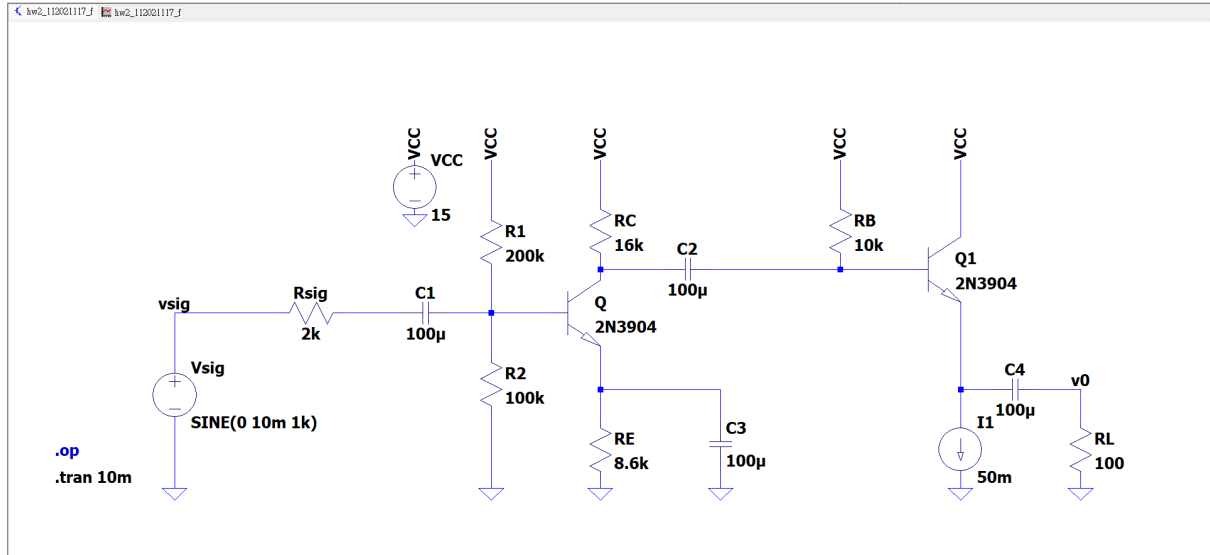


Figure 14: Complete schematic of the CE amplifier cascaded with the Emitter Follower stage.

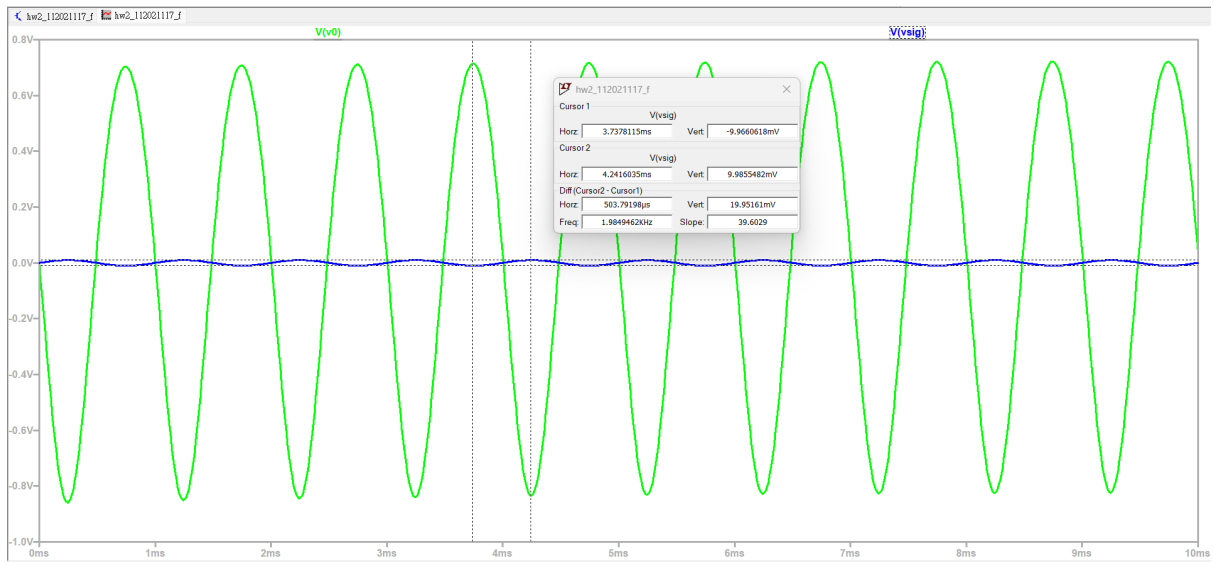


Figure 15: Simulated input waveform v_{sig} indicating $19.95mV_{pp}$.

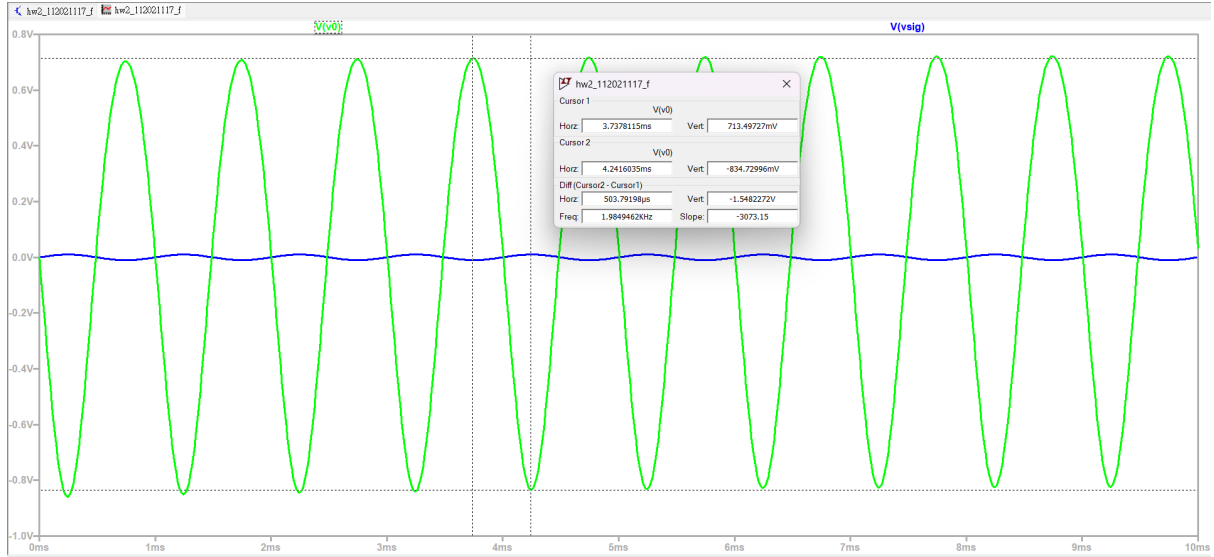


Figure 16: Undistorted output waveform v_o across the 100Ω load indicating $1.548V_{pp}$.

5. Verification of Non-Ideal R_B Choices

Two non-ideal cases were simulated to verify the consequences of violating the derived boundaries:

- **Under-limit Case ($R_B = 5k\Omega$):** With a 10mV input, the heavy loading effect reduces the output swing $V_{o,pp}$ to 1.046V (Figure 17), degrading the total gain to ≈ 52.43 . This confirms that a low R_B reduces the gain margin.
- **Over-limit Case ($R_B = 30k\Omega$):** With a large-signal 100mV input, the excessive DC voltage drop shifts the Q-point too low, causing severe asymmetric clipping near the negative peaks and restricting the negative swing boundary to $-2.81V$ (Figure 18).

Conclusion: These results validate that $R_B = 10k\Omega$ is the optimized choice to ensure both adequate gain margin and large-signal headroom.

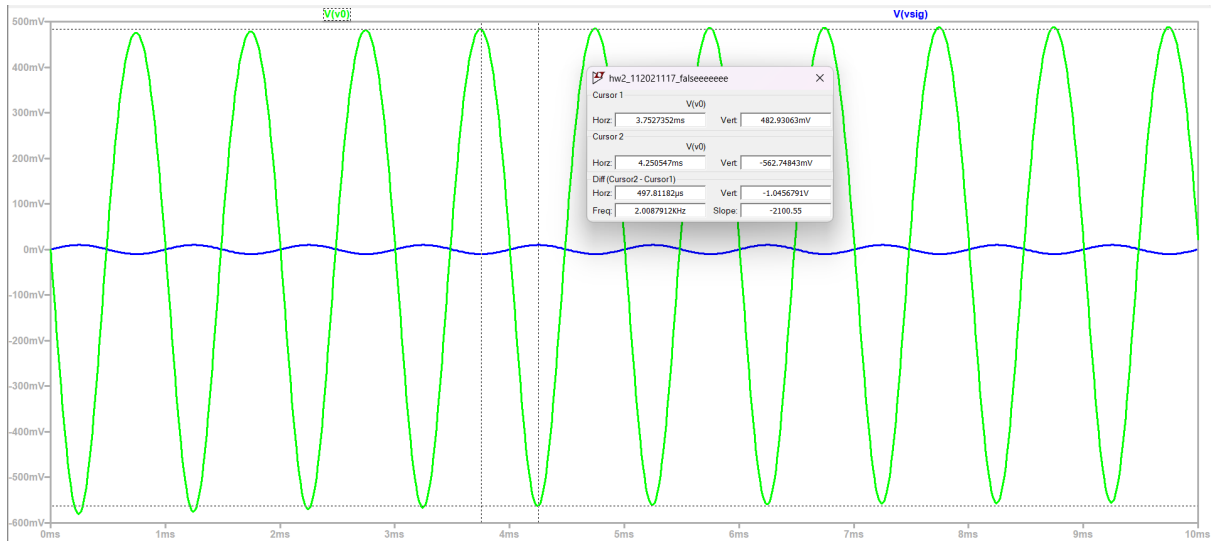


Figure 17: Output waveform v_o showing reduced swing ($1.046V_{pp}$) when $R_B = 5k\Omega$.

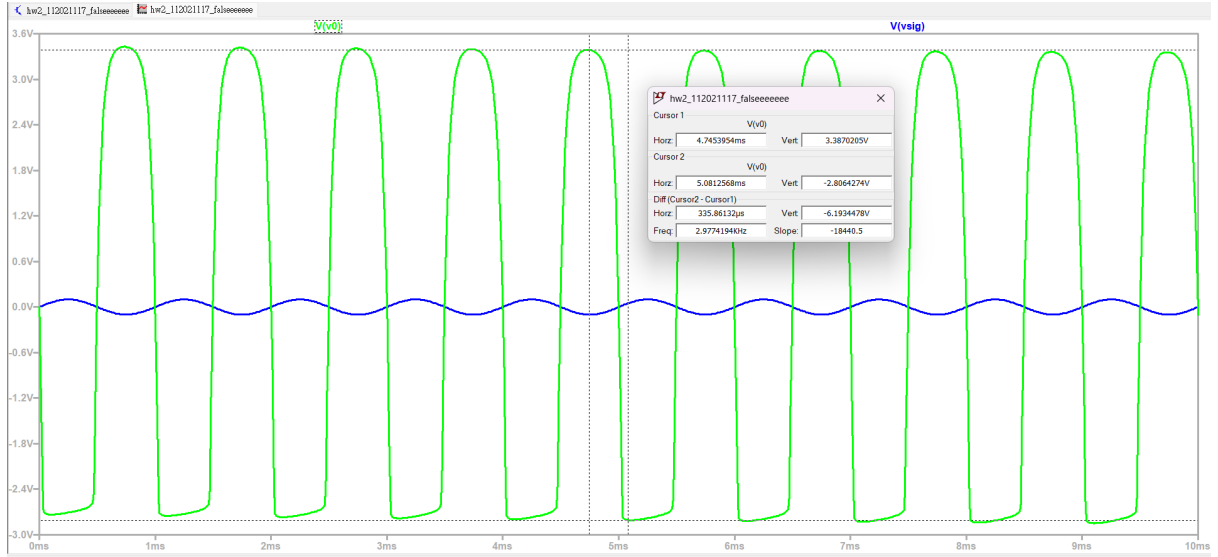


Figure 18: Asymmetric waveform distortion and clipping at 100mV input amplitude when $R_B = 30k\Omega$.